SkillGauge AI – Interview Assessment Report

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1. Which Python data structure is unordered, mutable, and defined by curly braces, storing elements as key-value pairs?
A) List
B) Tuple
C) Set
D) Dictionary

2. In C programming, which standard library function is used for dynamic memory allocation, returning a void pointer to the allocated space?
A) calloc()
B) realloc()
C) malloc()
D) free()

3. Which SQL clause is used to filter results based on aggregate functions (like COUNT or AVG)?
A) WHERE
B) GROUP BY
C) HAVING
D) ORDER BY

4. In HTML, which tag is semantically used to define self-contained content that should be independently distributable?
A) <section>
B) <aside>
C) <article>
D) <header>

5. Which CSS property is used to control the external spacing between an element's border and other adjacent elements?
A) Padding
B) Margin
C) Border-spacing
D) Outline

6. In the standard Machine Learning Pipeline, what crucial step typically occurs immediately after 'Feature Engineering' and before 'Model Training'?
A) Data Collection
B) Hyperparameter Tuning
C) Model Deployment
D) Data Splitting (Train/Test)

7. When evaluating a classification model, what metric is defined as (True Positives) / (True Positives + False Positives)?
A) Recall
B) Accuracy
C) Precision
D) F1-Score

8. What command in Git (used with GitHub) is used to integrate changes from one branch into the current branch?
A) git stash
B) git fetch
C) git merge
D) git checkout

9. VS Code allows developers to manage different sets of settings, extensions, and environment variables specific to a project. What is this feature called?
A) Workspaces
B) Profiles
C) Sessions
D) Code Groups

10. Dataiku is primarily a platform focused on facilitating which process?
A) Real-time stream processing only
B) Low-code application development
C) Collaborative Data Science and MLOps lifecycle management
D) NoSQL database management

11. TIBCO Spotfire is a widely used tool primarily specialized for which purpose in data analysis?
A) Database administration
B) Interactive data visualization and business intelligence dashboards
C) Server-side web scripting
D) Container orchestration

12. Google BigQuery is designed as a serverless data warehouse that separates storage and compute. Which standard SQL dialect does it use?
A) Transact-SQL
B) Standard SQL (2011)
C) PL/SQL
D) HiveQL

13. During data cleaning, if you identify a column where 80% of the values are missing, what is the most appropriate action if the column is not critical to the prediction target?
A) Impute using the mean of the remaining 20%
B) Perform multiple imputation
C) Drop the column entirely
D) Impute using a constant value like zero

14. In pattern detection, identifying cyclical variations that repeat consistently over a fixed period (e.g., weekly or annually) in time-series data is known as detecting:
A) Noise
B) Trend
C) Seasonality
D) Anomalies

15. Which Python micro-framework is specifically designed for building web applications and is often used for creating APIs and smaller web services?
A) Django
B) Flask
C) Pyramid
D) FastAPI

16. MongoDB is a NoSQL database that stores data in which flexible, schema-less format?
A) Relational Tables
B) Key-Value Pairs
C) Graph Nodes and Edges
D) BSON Documents

17. A machine learning model suffering from high variance is said to be doing what?
A) Underfitting (High Bias)
B) Overfitting (High Variance)
C) Stagnating (Low Error)
D) Balancing Bias and Variance

18. Which branch of Artificial Intelligence focuses on enabling computer systems to learn from data without explicit programming, leading to predictive models?
A) Robotics
B) Natural Language Processing
C) Machine Learning
D) Expert Systems

19. For the 'Suicide Rate Prediction And Analysis Using Machine Learning' project, if the target variable is the continuous numerical value of 'rate per 100k population', which type of modeling is required?
A) Clustering
B) Classification
C) Regression
D) Dimensionality Reduction

20. In a 'Bank Customer Churn Prediction' project, which specific evaluation metric should be prioritized if the bank wants to ensure they correctly identify *most* customers who are actually planning to leave (minimizing False Negatives)?
A) Precision
B) Specificity
C) Recall
D) Mean Squared Error

21. The Xception deep learning architecture, often used for tasks like Diabetic Retinopathy prediction, is characterized by replacing standard convolution layers with what specific technique?
A) Recurrent Neural Networks
B) Residual Connections
C) Depthwise Separable Convolutions
D) Attention Mechanisms

22. The AI-Powered System for Glaucoma and Otitis Media Detection primarily falls under which subfield of AI?
A) Generative Adversarial Networks (GANs)
B) Reinforcement Learning
C) Computer Vision
D) Time-Series Forecasting

23. What is the term for parameters, such as the learning rate or the number of trees in a Random Forest, that are set externally before the model training begins?
A) Coefficients
B) Features
C) Hyperparameters
D) Weights

24. Predictive models are fundamentally designed to achieve which outcome?
A) Explain causation among variables
B) Forecast unknown future events or values
C) Provide definitive answers to counterfactual questions
D) Manipulate real-time data streams

25. Which data cleaning technique is most appropriate for handling categorical features that contain minor misspellings or inconsistent case (e.g., 'NY', 'New York', 'new york')?
A) One-Hot Encoding
B) Feature Scaling
C) String matching and standardization
D) Polynomial transformation

FINAL SCORE: 9/25